The Joy of Computing using Python

Assignment 1

1. Which of the following is true about a computer program?

   (a) It is a sequence of instructions

   (b) Instructions that are written in simple english

   (c) There is only one universal programming language

   (d) It is meant for only software developers

   CORRECT ANSWER: (a)

2. Assume you are given two images, each displaying two basic positions of situps. Identify the set of commands to perform an exercise with both images.

   (a) Turn clockwise 90 degree/Turn clockwise -90 degrees

   (b) Move 10 steps/Move -10 steps

   (c) Hide/Show

   (d) point in direction 90/point in direction -90

   CORRECT ANSWER: (c)

3. Choose the best command to be used at the start of your code, to locate the sprite at an initial position every time you play the animation.

   (a) change x to val

   (b) set x to val / set y to val

   (c) point in direction 90

   (d) point towards mouse pointer

   CORRECT ANSWER: (b)

4. Assume the sprite is a ball and predict the output of the following control structure.



   (a) The ball glides to a random position in 1 second

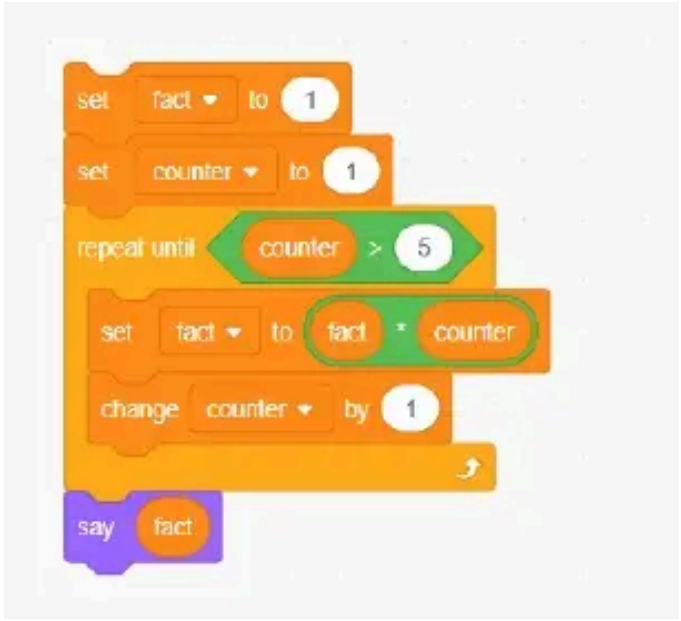
   (b) The ball glides to 10 random positions taking 1 second to reach each position

   (c) The ball glides to a random position and waits there for 10 seconds

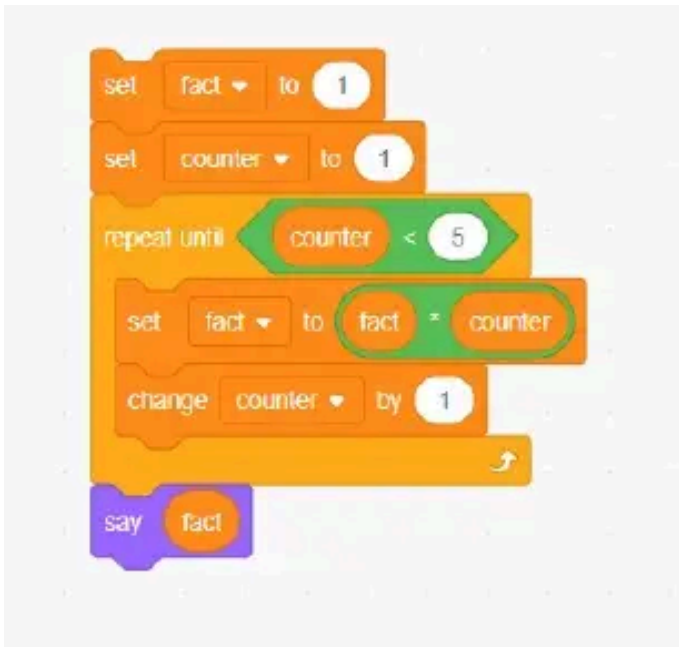
   (d) The ball glides to 10 random positions within 1 second

   CORRECT ANSWER: (b)

5. Pick the snippet that helps the sprite to find the factorial of 5 as output.
   Given:
   Factorial(n)=1 x 2 x 3 x .. x (n-1) x n



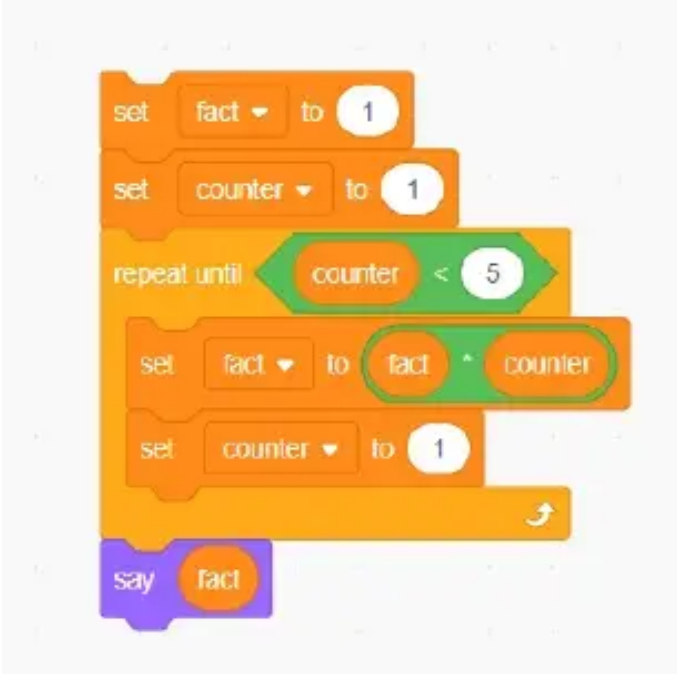
(a)



(b)



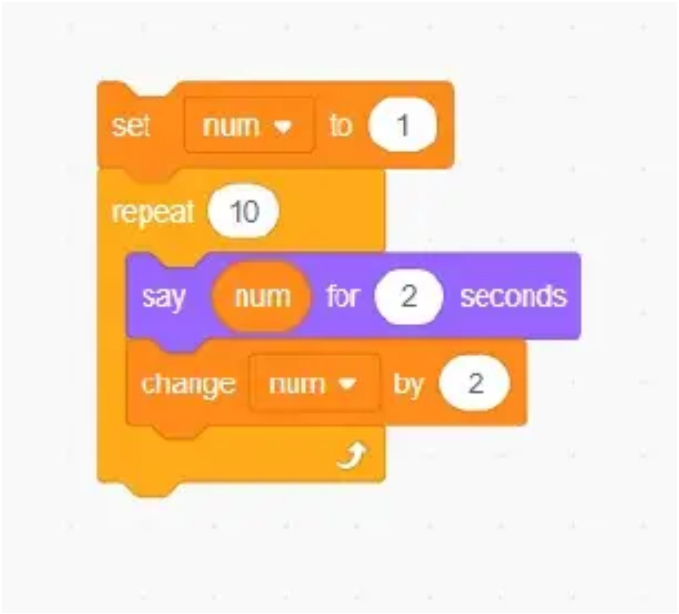
(c)



(d)

CORRECT ANSWER: (a)

6. Predict the sequence of numbers that the sprite recites:



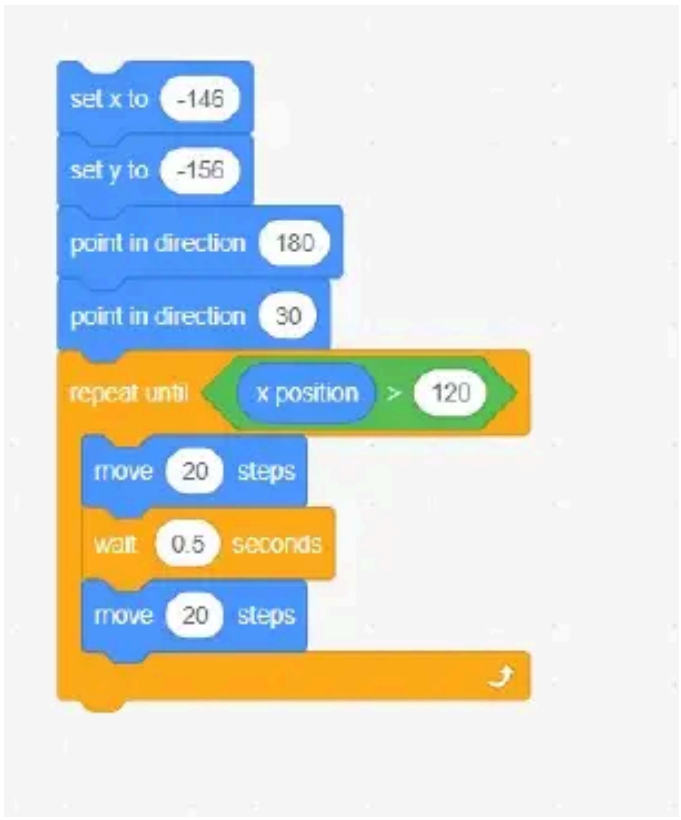
(a) 1, 2, 3, . . . , 19
(b) 1, 3, 5, . . . , 19
(c) 3, 5, 7, . . . , 19
(d) 1, 2, 3, . . . , 20

CORRECT ANSWER: (b)

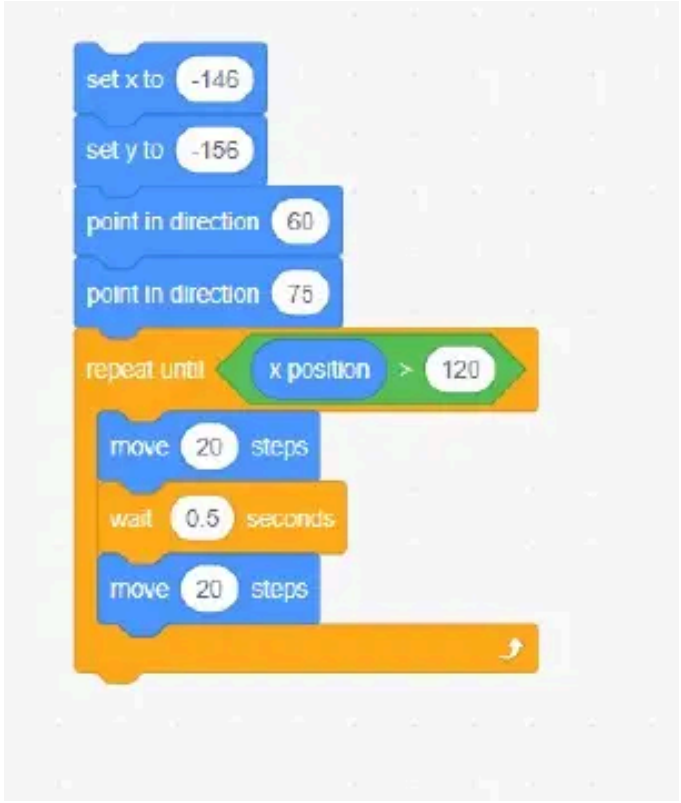
7. Consider a road inclined at an angle of 30° and we have a car sprite to be driven over this road. Pick the code that helps to perform the same. Hint: The initial direction of the sprite is 90°.



(a)



(b)



(c)



(d)

CORRECT ANSWER: (d)

8. Identify the command to communicate across multiple sprites.

   (a) say message
   (b) play sound
   (c) Broadcast message
   (d) touching color

   CORRECT ANSWER: (c)

9. Pick out the scratch library that provides the functionality to switch backdrop?

   (a) Motion

   (b) Control

   (c) Looks

   (d) Sensing

   CORRECT ANSWER: (c)

10. Imagine a Magic wand sprite and predict the output for the following set of instructions.



   (a) The Magic wand flies to a random position takes 1 sec pause and then reaches another random position.

   (b) The Magic wand flies to a random position in 1 sec and after a 0.25 sec pause, it repeats the same until it is stopped.

   (c) The Magic wand reaches all edges of the screen in a uniform pattern

   (d) The Magic wand flies between the top and bottom edges repeatedly.

   CORRECT ANSWER: (b)

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Assignment 2

NOTE: Python 3.7 has been used for this Assignment

1. Identify the statement with an invalid syntax. Expected Output: Stay Safe! Friends

   (a) print("Stay Safe!","Friends")

   (b) print("Stay Safe! Friends")

   (c) print('Stay Safe! Friends')

   (d) print(Stay Safe Friends)

   CORRECT ANSWER: (d)

2. Predict the output for the following code that checks the eligibility to vote.

```
age=19
if age>=18
        print('Hey! You are eligible to vote')
else:
        print('OOPS! You are not eligible to vote')
```

   (a) Hey! You are eligible to vote

   (b) OOPS! You are not eligible to vote

   (c) Syntax Error: invalid syntax

   (d) Invalid age

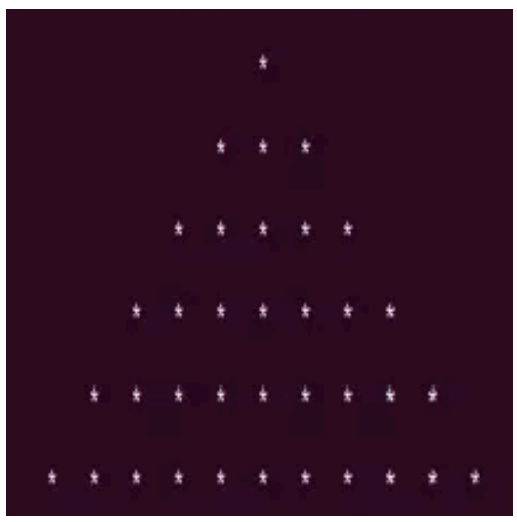
   CORRECT ANSWER: (c)

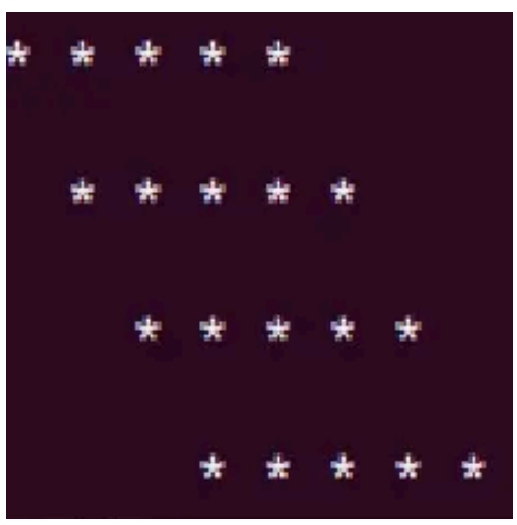
   The option Syntax Error: invalid syntax is correct because colon is missing in the if statement

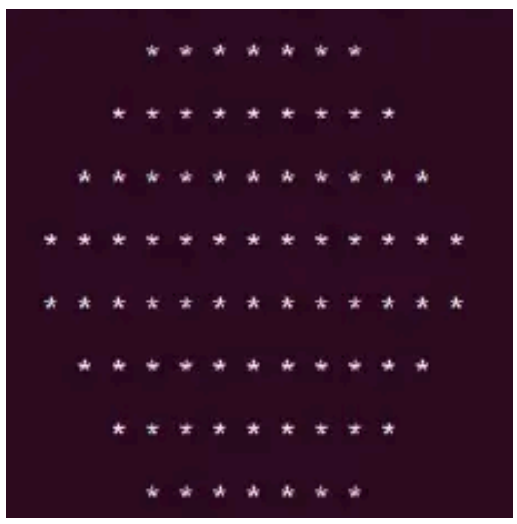
3. What is the output of the following code?

```
def func():
        print()
        c=10
        i=0
        while(i<=5):
                j=1
                while(j<=20):
                        print(' ',end=' ')
                        if(j>=10-i and j<=10+i):
                                print('*',end=" ")
                        else:
                                print(' ',end=" ")
                        j=j+1
                print('\n')
                i=i+1
func()
```

(a) 

(b) 

(c) 

(d) 

CORRECT ANSWER: (b)

4. With n as input, the code below computes

```
def mul(num):
        if(num==1):
                return(-1)
        return(-1*mul(num-1))
```

```
n=int(input("Enter the value of n"))
print(mul(n))
```

(a) $-1 \times n$

(b) $-1 + n$

(c) $(-1)^n$

(d) $n^{(-1)}$

CORRECT ANSWER: (c)

5. For the given code, what is the value of 'total' variable at the end of execution?

```
m1 = '98'
m2 = '79'
m3 = '87'
total = m1 + m2 + m3
print(total)
```

(a) syntax error

(b) 264

(c) 988779

(d) 987987

CORRECT ANSWER: (d)

The last option is correct because the operator '+' concatenates the string inputs.

6. Replace the given set of instructions with a for loop.

```
n=2
print(n)
n=n*2
print(n)
n=n*2
print(n)
n=n*2
print(n)
n=n*2
print(n)
```

(a)
```
for i in range(1,6):
print(pow(2,i))
```

(b)
```
for i in range(1,6):
    print(pow(i,2))
```

(c)
```
for i in range(1,6):
    print(pow(2,i))
```

(d)
```
for i in range(1,5):
    print(pow(2,i))
```

CORRECT ANSWER: (c)

7. Which of the following is the output for the given code?

```
n=5;print(n+5);print(n+5);print(n+5);print(n+5);
```

(a) 10 10 10 10

(b) 5 10 15 20

(c) 10
10
10
10

(d) 5
10
15
20

CORRECT ANSWER: (c)

8. Identify the appropriate output.

```
name='Avani Chaturvedi!'
print('Hello!', name, 'How are you?')
print('Proud to meet you!')
```

(a) Hello! Avani Chaturvedi! How are you? Proud to meet you!

(b) Hello! Avani Chaturvedi!
How are you?
Proud to meet you!

(c) Hello!
Avani Chaturvedi!
How are you?
Proud to meet you!

(d) Hello! Avani Chaturvedi! How are you?
Proud to meet you!

CORRECT ANSWER: (d)

9. What happens if we key in number 5 for the variable c in the below code?

```
c=1
while(c==1):
        print('hello')
        c=int(input('Enter choice:0/1:'))
```

—

(a) Loop terminates

(b) Continues execution

(c) Program restarts execution

(d) None of the above

CORRECT ANSWER: (a)

Loop terminates is the correct answer because the loop continues only when c is equal to 1. For all other values it terminates.

10. What is the output of the following code?

```
d=-20
while(d>0):
    print(d)
if(d<0):
    print(-1*d)
```

(a) Infinite loop

(b) -20 20

(c) 20

(d) -20

CORRECT ANSWER: (c)

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Assignment 3

NOTE: Python 3.7 has been used for this Assignment

1. What is the expected output for the following code?

```
cart=['coffee','sugar','cheese','butter']
for item in cart:
        if item=='sugar':
                print('jaggery')
        else:
                print(item)
```

   (a) ['coffee', 'jaggery', 'cheese ', 'butter ']

   (b) ['coffee ', 'sugar ', 'cheese ', 'butter ']

   (c) coffee
   jaggery
   cheese
   butter

   (d) coffee jaggery cheese butter

   CORRECT ANSWER: (c)

   Solution: The items in the list named 'cart' are only displayed, not the list itself. So the first two options are wrong. The print statement adds by default a newline character at the end. Therefore only option three is right.

2. Which of the following code prints the sum of weights of people in the lift.

   (a)
```
sum=0
weights=[97, 52, 65, 43, 77]
for w in weights:
        sum=sum+w
print(sum)
```

   (b)
```
sum=0
weights=[97, 52, 65, 43, 77]
for w in range(len(weights)):
        sum=sum+w
print(sum)
```

   (c)
```
sum=0
weights=[97, 52, 65, 43, 77]
for w in weights:
sum=sum+w
print(sum)
```

   (d)
```
sum=0
weights=[97, 52, 65, 43, 77]
for w in weights:
        sum=w
print(sum)
```

CORRECT ANSWER: (a)

First option is correct. Indenting is wrong in the second and third options whereas in the third option, the sum is not calculated by adding the items in the list in the last option.

3. Consider a python list named 'book_titles'. Pick the statement to add 'Who moved my cheese?' as the third item.

   Given: book_titles = ['Exam Warriors', 'Evil in the Mahabharata', '6 TIMES THINNER', 'The Driver in the Driverless Car', 'Evolution']

   (a) book_titles.append(2,'Who moved my cheese?')
   (b) book_titles.insert(2,'Who moved my cheese?')
   (c) book_titles.insert(3,'Who moved my cheese?')
   (d) book_titles.append(3,'Who moved my cheese?')

   CORRECT ANSWER: (b)

   Solution: book_titles.insert(2,'Who moved my cheese?') is correct.

   insert() function is used to add an item to the list at a desired index position. The indexing starts from 0. Therefore option two is correct.

4. Pick the relevant output for the given code.

```
n=[1 ,4 ,2 ,8 ,21 ,17]
n. reverse ()
print(n)
```

   (a) [1, 2, 4, 8, 17, 21]
   (b) [21, 17, 8, 4, 2, 1]
   (c) [17, 21, 8, 2, 4, 1]
   (d) [1, 4, 2, 8, 21, 17]

   CORRECT ANSWER: (c)

   Solution: [17, 21, 8, 2, 4, 1] is correct. The reverse() function exactly reverses the sequence of elements in the list.

5. Specify the purpose of 'break' statement inside a nested loop.

   (a) Ends execution of the program
   (b) Ends execution of the outermost loop
   (c) Skips the current iteration of the loop
   (d) Ends the execution of the loop

   CORRECT ANSWER: (d)

6. You are given a list, 'marks' scored by 30 students. Identify the instruction to find the 2% trimmed mean for the given data.

   (a) m=stats.trim_mean(marks,0.2)
   (b) m=stats.trim_mean(marks,0.03)
   (c) m=stats.trim_mean(30,0.02)
   (d) m=stats.trim_mean(marks,0.02)

   CORRECT ANSWER: (d)

7. How will you simulate 'Rolling a Dice' with six faces by making use of 'random' library?

   (a) roll= random.choice(1,2,3,4,5,6)
   (b) roll= random.range(1,5)
   (c) roll= random.randint(1,6)

(d) roll= random.random(6)

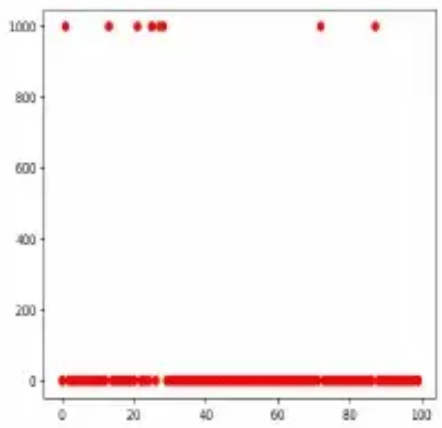
CORRECT ANSWER: (c)

8. Which of the plots in the options is most likely to be generated from the following code?

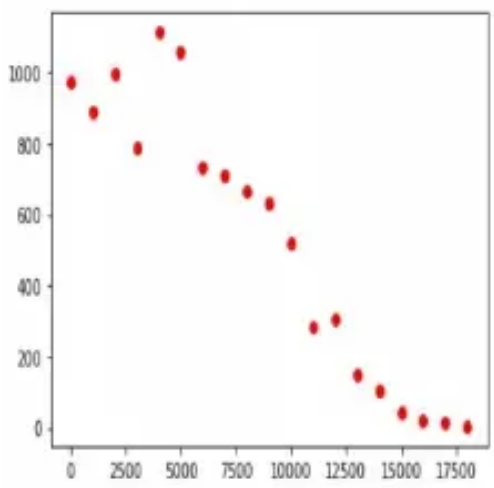
```
import random
import matplotlib.pyplot as plt

def play():
        amt=0
        for i in range(0,100):
                r=random.randint(1,1000)
                if(r!=random.randint(1,1000)):
                        amt=amt
                else:
                        amt=amt+1000
        return amt

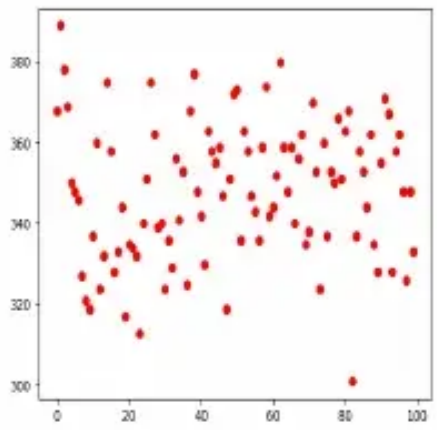
l=[]
for j in range(0,100):
        s=0
        for i in range(0,100):
                s=s+play()
        l.append(s)
x=[]
y=[]
for each in list(set(l)):
        x.append(each)
        y.append(l.count(each))
plt.plot(x,y,'ro')
plt.show()
```



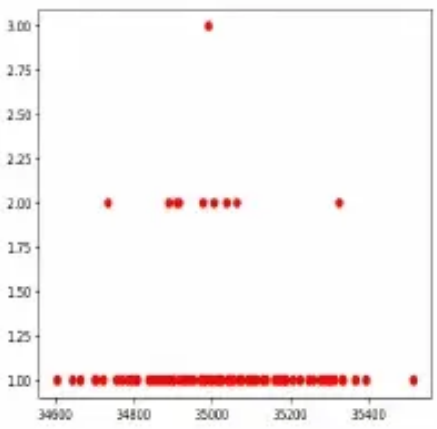
(a)



(b)



(c)



(d)

CORRECT ANSWER: (b)

9. Assuming, there is no file named 'file.txt' on my computer, what does the following code do?

```
with open('file.txt','w') as f:
        f.write('Hey! I am writing.');
f.close()
with open('file.txt','w') as f:
        f.write('Hey I am writing the second line.');
f.close()
with open('file.txt','r') as f:
        print(f.read())
f.close()
```

(a) Shows error

(b) Displays: Hey I am writing the second line

(c) Displays: Hey! I am writing.Hey I am writing the second line.

(d) Displays: Hey! I am writing.

CORRECT ANSWER: (b)

10. Predict the output

```
my_para='i am to go to KT in A'
print(list(my_para))
```

(a) `['i', ' ', 'a', 'm', ' ', 't', 'o', ' ', 'g', 'o', ' ', 't', 'o', ' ', 'K', 'T', ' ', 'i', 'n', ' ', 'A']`

(b) `['i', 'a', 'm', 't', 'o', 'g', 'o', 't', 'o', 'K', 'T', 'i', 'n', 'A']`

(c) `['i', 'am', 'to', 'go', 'to', 'KT', 'in', 'A']`

(d) `['i', ' ', 'am',' ', 'to',' ', 'go',' ', 'to',' ', 'KT',' ', 'in',' ', 'A']`

CORRECT ANSWER: (a)

First option is correct because every element including whitespace in the input string becomes an element in the output list.

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Assignment 4

NOTE: Python 3.7 has been used for this Assignment

1. Which statement can be used to come out of an infinite loop?

   (a) continue
   (b) break
   (c) try
   (d) catch

   CORRECT ANSWER: (b)

2. You are supposed to code your 'To do' list that contains all the activities that you plan to perform in a day. Assume you recharged your mobile and want to delete it from the list. Identify the statement to perform the same.

   Given: to-do=['Send Email', 'Recharge Mobile', 'Workshop preparation']

   (a) to-do.delete("Recharge Mobile ")
   (b) to-do.clear("Recharge Mobile")
   (c) to-do.remove("Recharge Mobile")
   (d) to-do.pop()

   CORRECT ANSWER: (c)

   Third option is correct. There is no delete() function for a list. clear() function takes no argument and it deletes all elements of the list, pop() function returns the last element of the list.

3. Simulate a 'Lot Box' that contains all alphabets from 'A' to 'Z'. Draw and Display.

   (a) print(random.choice(list(string.ascii_letters)))
   (b) print(random.choice(list(string.ascii_uppercase)))
   (c) print(random.choice(list(string.ascii_lowercase)))
   (d) print(random.choice(string.ascii_uppercase))

   CORRECT ANSWER: (b)

4. The following snippet produces TypeError: int object not callable. Pick out the correct code.

   result=a(b+(c**2)) where a, b **and** c are **any** integers

   (a) result=ax(b+(c**2))
   (b) result=a//(b+(c**2))
   (c) result=a*(b+(c**2))
   (d) result=a.(b+(c**2))

   CORRECT ANSWER: (c)

5. How will you display the current date in 'mm/dd/yy' format?

   (a) print(datetime.datetime.now().strftime('%c'))
   (b) print(datetime.datetime.now().strftime('%B'))

(c) print(datetime.datetime.now().strftime('%C'))

(d) print(datetime.datetime.now().strftime('%x'))

CORRECT ANSWER: (d)

6. What does the following code do?

```
s1=input('Enter a string')
s2=input('Enter another string')
for each in list(s2):
        for each2 in list(s1):
                if(each==each2):
                        print('yes')
                        break
```

(a) prints yes if both strings are same

(b) prints yes if both strings have atleast one common character

(c) prints yes if first string is contained in the second

(d) none of the above

CORRECT ANSWER: (b)

7. What does the following function do?

```
def leap(year):
        if(year % 400 == 0 or (year % 100 != 0 and year % 4 == 0)):
                return 1
        else:
                return 0
```

(a) returns true for century year and false for non century year

(b) returns true for leap year and false for non leap year

(c) returns false for century year and true for non century year

(d) none of the above

CORRECT ANSWER: (b)

8. Given a $n \times n$ square matrix mx in the form of list of lists in figure, what is the output of the statement func(mx,2) given n=7?

1	2	3	4	5	6	7
8	9	0	9	8	7	6
5	4	3	2	1	2	3
4	5	6	7	8	9	0
9	8	7	6	5	4	3
2	1	2	3	4	5	6
7	8	9	0	9	8	7

```
def func(mx, i):
        for ind in range(i,n-i):
                print(mx[i][ind],end=' ')
        for ind in range(i+1,n-i):
                print(mx[ind][n-1-i],end=' ')
        for ind in range(n-2-i,i,-1):
                print(mx[n-1-i][ind],end=' ')
        for ind in range(n-i-1,i,-1):
                print(mx[ind][i],end=' ')
```

(a) 3 2 1 8 5 6 7 6

(b) 3 2 1 6 7 8 7 6 5

(c) 3 2 1 1 8 5 5 6 7 7 6 3

(d) 3 6 7 7 7 2 1 8 5

CORRECT ANSWER: (a)

9. Pick out the snippet to perform integer division.

(a) a // b
(b) a / b
(c) a mod b
(d) a % b

CORRECT ANSWER: (a)

10. Pick out the valid function call to the definition given below:

```
def is_participating(name, participants):
        c=participants.count(name)
        if c==0:
                return(False)
        else:
                return(True)
```

(a) is_participating('Raji','Shiva','Raji','Priya')
(b) is_participating('Raji',['Shiva','Raji','Priya'])
(c) is_participating['Raji',{'Shiva','Vani','Priya'}]
(d) is_participating('Raji',{'Shiva','Raji','Priya'})

CORRECT ANSWER: (b)

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Assignment 5

NOTE: Python 3.7 has been used for this Assignment

1. Select the command to empty or reset the 'employee' dictionary.

   (a) del employee

   (b) del employee[0:2]

   (c) employee.remove()

   (d) employee.clear()

   CORRECT ANSWER: (d)

2. Which of the following code represents creating a dictionary from a list where keys are the unique elements from the list and the value corresponding to a key is the number of times that key occurs in the list.

   (a)
```
dict1={}
list1=[1,2,4,5,3,2,4,5,6,7,8,1,2,3,4,6,9,10]
for each in list1:
        if each not in dict1:
                dict1[each]=1
        else:
                dict1[each]=dict1[each]+list1.count(each)
print(dict1)
```

   (b)
```
dict1={}
list1=[1,2,4,5,3,2,4,5,6,7,8,1,2,3,4,6,9,10]
for each in list1:
        if each not in dict1:
                dict1[each]=0
        else:
                dict1[each]=dict1[each]+list1.count(each)
print(dict1)
```

   (c)
```
dict1={}
list1=[1,2,4,5,3,2,4,5,6,7,8,1,2,3,4,6,9,10]
for each in list1:
        if each not in dict1:
                dict1[each]=1
        else:
                dict1[each]=dict1[each]+1
print(dict1)
```

   (d) none of these

   CORRECT ANSWER: (c)

3. Identify the audio file format that is NOT supported by Python Speech Recognition Module.

   (a) FLAC

   (b) AIFF

   (c) WAV

   (d) MP3

CORRECT ANSWER: (d)

4. Which of the following exception can be used to handle the error that occurs when Google cannot understand the audio content in speech recognition?

   (a) UnknownValueError
   (b) RequestError
   (c) ValueError
   (d) RunTimeError

   CORRECT ANSWER: (a)

5. Which of the following statements is correct for the Monte Hall problem?

   Statement I: If you choose the correct door on the first try, then switching loses

   Statement II: Contestants who switch have 2/3 chances to win whereas contestants who donot switch have 1/3 chances of win.

   (a) I only
   (b) II only
   (c) Both I & II
   (d) None

   CORRECT ANSWER: (c)

6. Which of the random experiments from the options does the code represent?

```
import random
p1=['rock','paper','scissor']
p2=['rock','paper','scissor']
c1=random.choice(p1)
c2=random.choice(p2)
if(c1==c2):
        print('SUCCESS')
else:
        print('FAIL')
```

   (a) Prints a success when both people select the same object
   (b) Prints a success when both people select “rock”
   (c) Prints a success when both people select different objects
   (d) None of the above

   CORRECT ANSWER: (a)

7. What does the following code represent?

```
import random
x=0
y=0
while(1):
        r = random.uniform(0,1)
        if (r<0.4):
                x=x+1
        elif (r<0.8):
                y=y+1
        else:
                x=x+1
                y=y+1
        print 'location=(',x,',',y,')'
        input("enter a key to continue")
```

(a) A drunkard moving on a straight line, moving one step forward with probability 0.4, one step backward with probability 0.4 and staying at the same place with probability 0.2

(b) A drunkard moving on a XY plane, moving right with probability 0.4, upwards with probability 0.8 and diagonally up-right with probability 1.

(c) A drunkard moving on a XY plane, moving right with probability 0.4, upwards with probability 0.4 and diagonally up-right with probability 0.2.

(d) A drunkard moving on a XY plane, moving left with probability 0.4, downwards with probability 0.8 and diagonally down-left with probability 1.

CORRECT ANSWER: (c)

8. The following code takes a list as input and prints the sorted list as an output. The outer for loop is to count the number of iterations. What is the purpose of the inner for loop?

```
def bubble(mylist):
        n=len(mylist)
        for i in range(n):
                for j in range(0,n-i-1):
                        if mylist[j]>mylist[j+1]:
                                mylist[j],mylist[j+1]=mylist[j
                                    +1] ,mylist[j]
        print(mylist)
```

(a) To fetch the pair of consecutive elements to be compared

(b) Index of the element for which the right position is to be found

(c) To identify the max element

(d) To check if the list is sorted

CORRECT ANSWER: (a)

9. The following code to its best, represents a scenario:

```
def func(i,f):
print(i)
if(i==0):
        f=1
        func(i+1,f)
if(i==128):
        f=-1
        func(i-1,f)
if(f==1):
        func(i+1,f)
if(f==-1):
        func(i-1,f)
```

(a) A cake getting eaten by half of its current amount every time

(b) A student attempting alternate questions, starting from a given question

(c) Viruses doubling inside a body and killing the person once their population becomes 128 or more.

(d) Metro train serving 128 stations to and fro

CORRECT ANSWER: (D)

10. Given that you have a sorted list of 1000 elements and the element to find is at the end of your list(worst case), what is the number of comparisons to search such an element using linear search and binary search?

(a) 1000, 10

(b) 10, 2

(c) 1000, 2

(d) 10, 10

CORRECT ANSWER: (a)

Solution: If n is the number of elements in the list, then it takes n comparisons for linear search and $log_2(n)$ comparisons for binary search

The Joy of Computing using Python

Assignment 6

NOTE: Python 3.7 has been used for this Assignment

1. Look at the following functions.

```
import random
import string

def create_encryption_key(string1):
    chars=list(set(list(string1)))
    keydict={}
    taken =[]
    for each in chars:
        while(1):
            r=random.choice(chars)
            if(r not in taken) :
                keydict[each]= r
                taken.append(r)
                break
    return(keydict)

def reverse(d):
    d1={}
    for each in d :
        d1[d[each]]= each
    return d1

def encrypt(letter,key):
    l=[]
    for i in range(0,len(letter)):
        l.append(key[letter[i]])
    return(l)
```

Which of the following set of statements correctly represent encryption and decryption using substitution cipher?

It is also given that the set of characters for substitution is chosen from the plain text.

(a)
```
plain_text=input("Enter the string you want to encrypt")
key=create_encryption_key(plain_text)
cipher_list= encrypt(plain_text,key)
cipher_text=(‘’.join(cipher_list))
plain_list= encrypt(cipher_list,reverse(key))
plain_text= (‘’.join(plain_list))
```

(b)
```
plain_text=input("Enter the string you want to encrypt")
key=create_encryption_key(plain_text)
cipher_list= encrypt(plain_text,key)
cipher_text=(‘’.join(cipher_list))
plain_list= encrypt(cipher_list,key)
plain_text= (‘’.join(plain_list))
```

(c)
```
plain_text=input("Enter the string you want to encrypt")
key=create_encryption_key(plain_text)
cipher_list= encrypt(plain_text,key)
```

```
cipher_text=(‘’.join(cipher_list))
plain_list= encrypt(plain_list,reverse(key))
plain_text= (‘’.join(plain_list))
```

(d) None of the above

CORRECT ANSWER: (a)

2. Assuming, there is no file named ”file.txt” on my computer, what does the following code do?

```
with open('file.txt','w') as f:
    print(f.read())
        f.write('Hey! I am writing');
f.close()
```

(a) Creates a file named file.txt and adds ‘Hey! I am writing’ to it

(b) Shows an error because file does not exist

(c) shows an error because file in not opened in the reading mode

(d) None of the above

CORRECT ANSWER: (d)

3. What does the function ’confidential’ do?

```
def confidential(mob_num):
        subs_dict={}
        sec_num=[0] * len(mob_num)
        for i in range(len(string.digits)):
                subs_dict[string.digits[i]]=string.digits[i-1]
        for j in range(len(mob_num)):
                sec_num[j]=subs_dict[mob_num[j]]
        return(sec_num)
```

(a) Generates the secret code for the given mobile number with every digit coded with the next digit.

(b) Generates the secret code for the given mobile number with every digit coded with the previous digit.

(c) Generates the secret code for the given mobile number with every digit coded with a random digit.

(d) Generates the secret code for the given mobile number with every digit coded with a special character.

CORRECT ANSWER: (b)

4. What is the output for the given code?

```
import numpy
mat=numpy.array([[1,2,3],[4,5,6],[7,8,9]])

def add(mat):
        sum=0
        for i in range(2):
                for j in range(2):
                        if i==j:
                                sum=sum+mat[i][j]
        return(sum)

print(add(mat))
```

(a) 15

(b) 9

(c) 6

(d) 24

CORRECT ANSWER: (c)

Solution: The code calculates the sum of all the diagonal elements of the numpy array whose indices i and j are equal.

5. Which of the following can be used to see the dimension of a numpy array names 'arr'?

   (a) dim(arr)
   (b) shape(arr)
   (c) arr.shape
   (d) arr.shape()

   CORRECT ANSWER: (c)

   Solution:

   The shape property is used to get the current shape of an array

6. What happens if we fail to check the anchor case in a recursive function?

   (a) Results in an infinite loop
   (b) RunTimeError
   (c) Never gets executed
   (d) Returns a wrong output

   CORRECT ANSWER: (a)

   Solution: The anchor case or base case is the condition that terminates the recursive call to the function. Therefore absence of it may cause an infinite loop.

7. What is the output of the following code ?

```
print('ab'.isalpha())
```

   (a) True
   (b) False
   (c) None
   (d) Error

   CORRECT ANSWER: (a)

8. If GOLD is encoded as FNKC, then how is PLATINUM encoded?,

   (a) NKYRGLSK
   (b) OKZSHMUL
   (c) NJYRGLSK
   (d) OKZSHMTL

   CORRECT ANSWER: (d)

   Solution: Each letter in GOLD is coded by its corresponding previous letter, FNKC. Therefore PLATINUM is coded as OKZSHMTL.

9. Which of these statements is true?

   (a) Recursion can solve only a subset of problems which Iteration can.
   (b) Recursion is not related to Iteration.
   (c) Recursion cannot solve the problems that can be solved by iteration.
   (d) Any problem that Recursion can solve, can also be solved by Iteration.

   CORRECT ANSWER: (d)

10. Which of the following strategy of play does Tic Tac Toe belong to?

    (a) Max-max

(b) Min-max

(c) Max-min

(d) Min-min

CORRECT ANSWER: (b)

Solution: Min- Max strategy which means to minimizing the chances of opponent's victory and maximizing one's own chances of achieving the winning configuration.

The Joy of Computing using Python

Assignment 7

NOTE: Python 3.7 has been used for this Assignment

1. Imagine a single player snakes and ladders game. The code below represents

```
import random

def play(psn, flag):
        snake_begin=-1
        snake_end=-1
        while(snake_begin <= snake_end):
                snake_begin=random.randint(1,99)
                snake_end=random.randint(1,99)
        print('Snake from',snake_begin,'to',snake_end)
        r = random.randint(1,6)
        print('Dice rolled:',r)
        if(psn==0):
                if(r==1 or r==6):
                        psn=1
        else:
                psn=psn+r
        print('Position=',psn)
        #input()
        if(psn==snake_begin and flag==0):
                print('Bitten by snake')
                psn=snake_end
                flag=1
        if(psn>=100):
                print('You won')
                return
        play(psn,flag)

position=0
print('Position=',position)
play(position,0)
```

   (a) A snakes and ladders game with one snake whose position remains constant while the player is playing. The position also remains the same during any subsequent plays (i.e. the game board does not change while you sleep and play again the next day).

   (b) A snakes and ladders game with one snake whose position remains constant while the player is playing. However, the position can change during any subsequent plays (i.e. the game board might change while you sleep and play again the next day).

   (c) A snakes and ladders game with one snake where the snake can change its position during the game and also during any subsequent plays (a board game where the snakes keep moving). Further, the snake can bite you any number of times.

   (d) A snakes and ladders game with one snake where the snake can change its position during the game and also during any subsequent plays (a board game where the snake keeps moving). Further, the snake can bite you only once when you play.

   CORRECT ANSWER : (d) The snake position is redefined each recursive call and there is flag mechanism to check if you have been bitten by a snake so that subsequent bites could be stopped.

2. Consider the code given below. Assume your current position being 19 and what happens when you roll 2?

```
import random
end=100

def snake_ladder(pos):
        sl_dict={1:38,4:14,8:30,21:42,28:76,32:10,
        36:6,48:26,50:67,62:18,71:92,80:99,88:24,95:56,97:78}
        if pos in sl_dict.keys():
                return(sl_dict[pos])
        else:
                return(pos)
def play():
        pos=0
        turn=0
        print('Let's play Snakes ans Ladders!')
        while(1):
                c=input('Press 1 to roll Dice/0 to Quit')
                if c=='0':
                        break
                roll=random.randint(1,6)
                turn=turn+1
                pos_roll=pos+roll
                pos=snake_ladder(pos_roll)
                if pos>pos_roll:
                        print('Hurray! You climbed a ladder.
                        Your score is ',pos)
                elif pos<pos_roll:
                        print('OOPS! You are bitten by a snake.
                         Your score is ',pos)
                else:
                        print('Your position is ', pos)
                if pos>=end:
                        print('You won in ',turn,' turns')
                        break

play()
```

(a) You climb up the ladder to reach 42

(b) You are bitten by a snake to reach 42

(c) You win

(d) You Quit

CORRECT ANSWER: (a)

3. Imagine a single player snakes and ladders game. The code below represents

```
import random

def play(psn):
        snake_begin=-1
        snake_end=-1
        while(snake_begin <= snake_end):
                snake_begin=random.randint(1,99)
                snake_end=random.randint(1,99)
        r = random.randint(1,6)
        print('Dice rolled:',r)
        if(psn==0):
                if(r==1 or r==6):
                        psn=1
        else:
                psn=psn+r
        if(psn==snake_begin):
```

```
                print('Bitten by snake')
                psn=snake_end
        if(psn>=100):
                print('You won')
                return
        play(psn)

position=0
play(position)
```

(a) A snakes and ladders game with one snake whose position remains constant while the player is playing. The position also remains the same during any subsequent plays (i.e. the game board does not change while you sleep and play again the next day).

(b) A snakes and ladders game with one snake whose position remains constant while the player is playing. However, the position can change during any subsequent plays (i.e. the game board might change while you sleep and play again the next day).

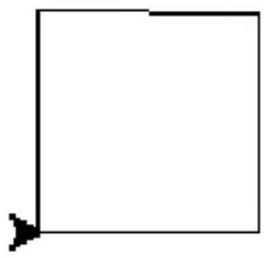
(c) A snakes and ladders game with one snake where the snake can change its position during the game and also during any subsequent plays (a board game where the snakes keep moving). Further, the snake can bite you any number of times.

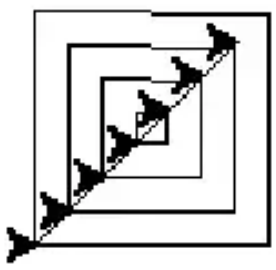
(d) A snakes and ladders game with one snake where the snake can change its position during the game and also during any subsequent plays (a board game where the snake keeps moving). Further, the snake can bite you only ones when you play.

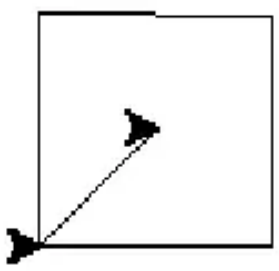
CORRECT ANSWER: (c) The snake position is redefined each recursive call and there is no mechanism to check if you have been bitten by a snake so that subsequent bites could be stopped.

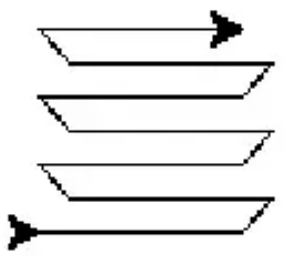
4. Predict the output of the calling function func(mx) for a given square matrix, mx of dimension 70×70.

```
def func(mx):
        f=1
        n=len(mx)
        tur = turtle.Turtle()
        tur.setpos(0,0)
        j=0
        while(j<n):
                if(f==1):
                        i=0
                        while(i<=n-1):
                                turtle.goto(i,j)
                                i=i+10
                if(f==0):
                        i=n-1
                        while(i>-1):
                                turtle.goto(i,j)
                                i=i-10
                f=(f+1)
                if(f==2):
                        f=0
                j=j+10
        turtle.done()
```

(a) 

(b) 

(c) 

(d) 

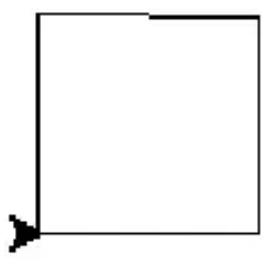
CORRECT ANSWER: (d)

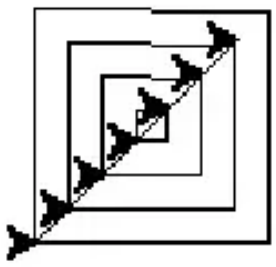
5. Predict the output of the calling function func() for a given square matrix mx of dimension 70×70.

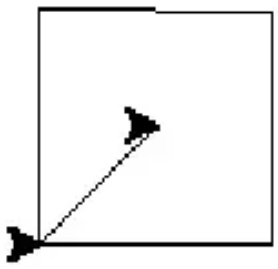
```
import turtle

def func():
    tur = turtle.Turtle()
    tur.setpos(0,0)
    n=len(mx)
    second=int(n/2)
    turtle.goto(second-1,second-1)
    turtle.goto(second-1,second)
    turtle.goto(second,second-1)
    turtle.goto(second,second)
    turtle.done()


func()
```

(a) 

(b) 

(c) 



(d)

CORRECT ANSWER: (d)

6. Identify the package that is used to import an image.

   (a) Pandas
   (b) Scipy
   (c) PIL
   (d) numpy

   CORRECT ANSWER: (c)

7. Given a $n \times n$ square matrix mx in the form of list of lists in the following figure, what is the output of the statement func(mx)?




```
def func1(mx, i):
        for ind in range(i,n-i):
            print(mx[i][ind],end=' ')
        for ind in range(i+1,n-i):
            print(mx[ind][n-1-i],end=' ')
        for ind in range(n-2-i,i,-1):
            print(mx[n-1-i][ind],end=' ')
        for ind in range(n-i-1,i,-1):
            print(mx[ind][i],end=' ')
```

```
def func(mx):
        for i in range(n):
            func1(mx, i)
            print()
```

(a) 3 2 1 8 5 6 7 6

(b) 1 2 3 4 5 6 7 6 3 0 3 6 7 8 9 0 9 8 7 2 9 4 5 8
9 0 9 8 7 2 9 4 5 4 3 2 1 8 5 4
3 2 1 8 5 6 7 6
7

(c) 1 2 3 4 5 6 7 6 3 0 3 6 7 8 9 0 9 8 7 2 9 4 5 8 9 0 9 8 7 2 9 4 5 4
3 2 1 8 5 4 3 2 1 8 5 6 7

(d) 3 6 7 7 7 2 1 8 5

CORRECT ANSWER: (b)

8. How do you create a base map using gmplot package?

(a) gmap=gmplot.GoogleMapPlotter(cent_lat,cent_long, zoom)

(b) gmap=gmplot.GoogleMapPlotter(cent_long,cent_lat, zoom)

(c) gmap=gmplot.GoogleMapPlotter(cent_lat,cent_long)

(d) gmap=gmplot.GoogleMapPlotter(zoom,cent_lat,cent_long)

CORRECT ANSWER: (a)

9. Comment on the following code:

```
import csv
from gmplot import gmplot

gmap=gmplot.GoogleMapPlotter(9.920227,78.158252,12)

gmap.coloricon = 'http://www.googlemapsmarkers.com/v1/%s'

with open('path1.csv','r') as d:
    reader = csv.reader(d)
    k=0

    for row in reader:
        lat=float(row[0])
        long=float(row[1])
        if(k==0):
            gmap.marker(lat,long,'red')
            k = 1
        else:
            gmap.marker(lat,long,'yellow')

gmap.draw('mymap.html')
```

(a) Red marker is used for initial and final positions

(b) Yellow marker is used for the final position only

(c) Yellow marker is placed for all positions except first position which has a red marker.

(d) Red marker is placed for all positions except first position which has a yellow marker

CORRECT ANSWER: (c)

10. How will you download and install packages that are unavailable in conda cloud?

(a) pip install packagename

(b) conda list

(c) conda install packagename

(d) sudo apt-get install packagename

CORRECT ANSWER: (a)

The Joy of Computing using Python

Assignment 8

NOTE: Python 3.7 has been used for this Assignment

1. What is the output of the following snippet?

```
t1=('Amit','Simran','Neeru','Ravi','Shubhadha')
t2=t1+t1
print(len(t2))
```

   (a) 5

   (b) 10

   (c) Error, because there is no len() function for Tuple

   (d) Error, because Tuples are immutable

   CORRECT ANSWER: (b)

   Solution: A tuple object can append to another tuple object

2. Which of the following instruction produces a tuple, Team with 'Poonam' as the sixth member?
   Given:

```
Team=('Amit','Simran','Neeru','Ravi','Shubhadha')
```

   (a) Team=Team.append('Poonam')

   (b) Team=Team+tuple('Poonam')

   (c) Team=Team+('Poonam')

   (d) Team=Team+('Poonam',)

   CORRECT ANSWER: (d)

3. Which of the scenarios in the options does the following code represent?

```
import random
def play():
        a=input('Enter a number from 1 to 10')
        r=random.randint(1,10)
        if(a==r):
                return 1
        else:
                return 0

amt=0
for i in range(1,366):
        amt=amt+play()

print(amt)
```

   (a) A person going to the bar for a year. Daily he guesses a number from 1 to 10. If the guessed number is equal to the number randomly generated by bar authority, he gains one gold coin.

   (b) A person going to the bar for a month. Daily he guesses a number from 1 to 10. If the guessed number is equal to the number randomly generated by bar authority, he gains one gold coin.

(c) A person going to the bar for a year. Daily he guesses a number from 1 to 10. If the guessed number is equal to the number randomly generated by bar authority, he loses one gold coin.

(d) A person going to the bar for a month. Daily he guesses a number from 1 to 10. If the guessed number is equal to the number randomly generated by bar authority, he loses one gold coin.

CORRECT ANSWER: (a)

4. Consider the following two codes:

Game 1

```
import random
r=random.uniform(0,1)
if(r<=0.5):
        print('won')
```

Game 2

```
import random
r=random.choice(range(1,6))
if(r%2==0):
        print('Won')
```

(a) Probability of winning in Game 1 > Probability of winning in Game 2.

(b) Probability of winning in Game 1 < Probability of winning in Game 2.

(c) Probability of winning in Game 1 = Probability of winning in Game 2.

(d) Can't say

CORRECT ANSWER: (a)

5. Choose the appropriate instruction to retrieve the mirror image of the given image.

(a) mirror_image=img.transpose(Image.FLIP_TOP_BOTTOM)

(b) mirror_image=img.flip(Image.FLIP_LEFT_RIGHT)

(c) mirror_image=img.transpose(Image.FLIP_LEFT_RIGHT)

(d) mirror_image=img.composite(Image.FLIP_LEFT_RIGHT)

CORRECT ANSWER: (c)

6. Identify the technique that can be used to enhance image in cv2.

(a) enh_img=clahe.apply(gray)

(b) enh_img=canny.apply(gray)

(c) enh_img=sobel.apply(gray)

(d) enh_img=enhance.apply(gray)

CORRECT ANSWER: (a)

7. Identify the best instruction to debug the following code that checks if the given strings are Anagrams:

```
s1=input('Enter first string:')

s2=input('Enter Second string:')
if s1.sort()!=s2.sort():
        print('These are Anagrams')
else:
        print('Not Anagrams')
```

(a) if s1.sort()==s2.sort():

(b) if s1.sorted()==s2.sorted():

(c) if sorted(s1)==sorted(s2):

(d) if sorted(s1)!=sorted(s2):

CORRECT ANSWER: (c)

8. Which of the following libraries helps us to find the intensity of emotion in sentiment analysis?

   (a) vader
   (b) nltk
   (c) pandas
   (d) scipy

   CORRECT ANSWER: (a)

9. The isalpha() function in NLTK

   (a) returns true if any of the words in a sentence are composed of alphabetic characters and false otherwise
   (b) returns true if all the characters in a word are alphabets and false otherwise
   (c) returns true if all the characters in a word are alphabets or numerics and false otherwise
   (d) None of the above

   CORRECT ANSWER: (b)

10. Every character, either alphabet or digit or special character has an ASCII value. Choose the appropriate method to find the ASCII value of 'f'.

    (a) ASCII('f')
    (b) ord('f')
    (c) int('f')
    (d) ASC_val('f')

    CORRECT ANSWER: (b)

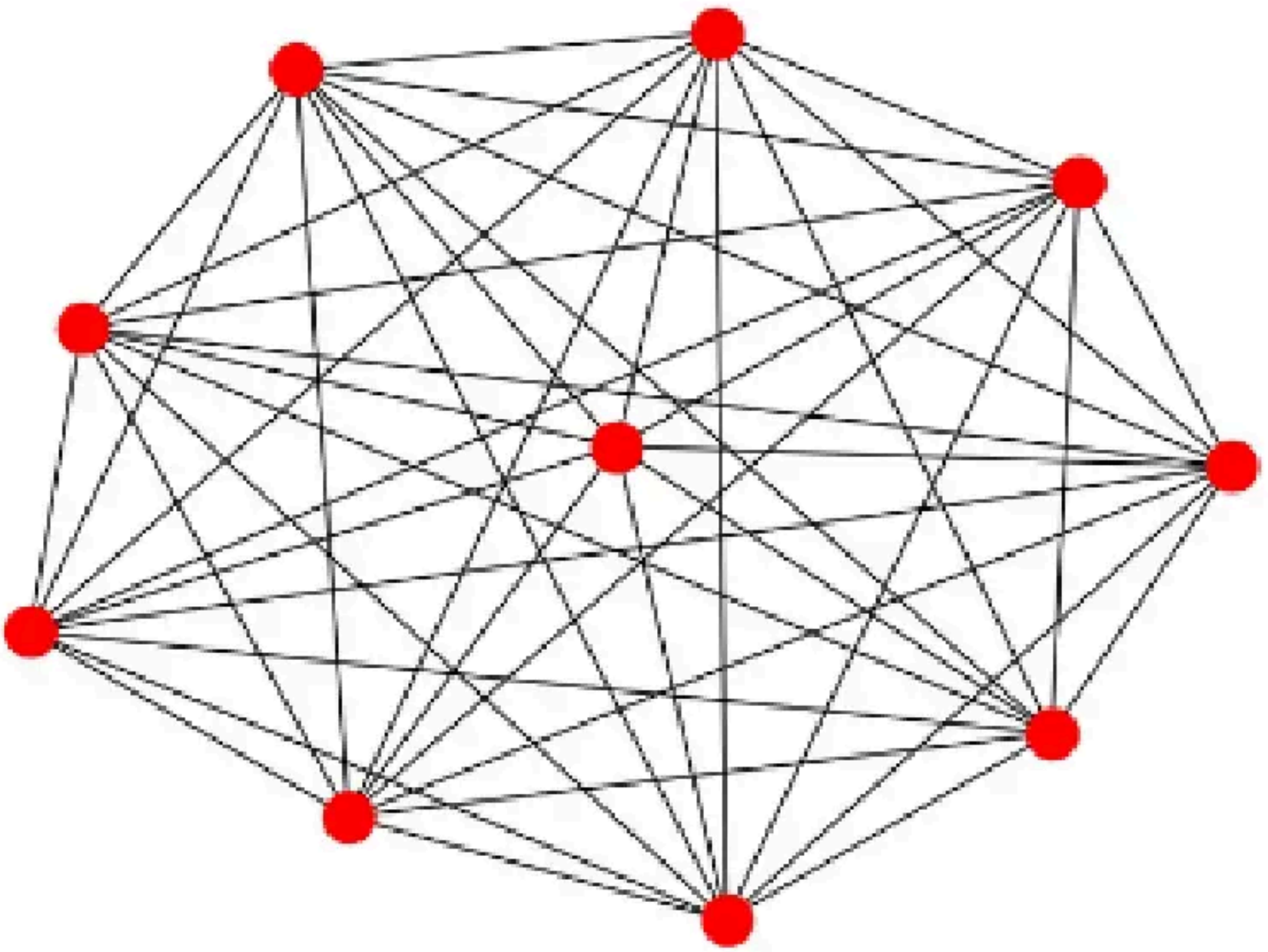
Assignment 9

Assessment due date is 18/11/2020 23:59:0

Last recorded submission : 2020-11-16, 13:28 IST

NOTE: Python 3.7 has been used for this Assignment

1) Which of the following commands is used to draw the following graph? ***1 point***



- G = nx.complete_graph(10)
- G = nx.complete_graph(9)

Assignment 9

- G = nx.Digraph(10)

Yes, The answer is correct
Score : 1

Accepted Answer

G = nx.complete_graph(10)

---

2) Identify the option that best describes the graph output of the following code. ***1 point***

```
import networkx as nx
import matplotlib.pyplot as plt

G = nx.Graph()
for i in range(1,11):
        G.add_node(i)
e = [(1,2),(2,3),(3,4),(4,5),(5,6),(6,7),(7,8),(8,9),(9,10)]
G.add_edges_from(e)

nx.draw(G)
plt.show()
```

- A complete graph with 10 nodes
- A complete graph with 9 nodes
- A connected graph with 10 nodes
- A connected graph with 9 nodes

Yes, The answer is correct
Score : 1

Accepted Answer

Assignment 9

3) How will you convert a graph, g into gexf format? ***1 point***

- nx.build_gexf(g,'Friendship.gexf')
- nx.create_gexf(g,'Friendship.gexf')
- nx.write(g,'Friendship.gexf')
- nx.write_gexf(g,'Friendship.gexf')

Yes, The answer is correct
Score : 1

Accepted Answer

nx.write_gexf(g,'Friendship.gexf')

4) What does p and q represent in the following instruction to find the shortest path length? ***1 point***

$$\text{networkx.shortest_path_length}(H, p, q)$$

- p and q represent to two lists of nodes
- p represents the source node and q represents the target node
- p represents the target node and q represents source the node
- p and q represent two paths

Yes, The answer is correct
Score : 1

Assignment 9

5) Given a newspaper article, the first step to perform text analytics is to break down the paragraph into smaller chunks of words. Select the instruction that does it for you. ***1 point***

- (●) word_tokenize(text)
- ( ) word_sent(text)
- ( ) text.word_tokenizer()
- ( ) text.sent_tokenizer()

Yes, The answer is correct
Score : 1

Accepted Answer

word_tokenize(text)

6) Predict the output for the given code: ***1 point***

- ( )
- (●) ['100', '%', '.', '.']
- ( )
- ( )

Assignment 9

Accepted Answer



7) Pick out the valid function to find the frequency distribution. ***1 point***

- nltk.FreqDist()
- nltk.Freq_Dist()
- nltk.Frequency_Dist()
- nltk.FDist()

Yes, The answer is correct
Score : 1

Accepted Answer

nltk.FreqDist()

8) Which of the following commands is used to create an image for an array? ***1 point***

- Image.array(array)
- Image.fromarray(array)

- Image.imgarray(array)

Yes, The answer is correct
Score : 1

Accepted Answer

Image.fromarray(array)

---

9) What is the output of the following snippet? ***1 point***

- Image of size 200 x 200 with Red on left and Blue on right
- Image of size 200 x 3 with Red on top and Blue at the bottom
- Image of size 200 x 200 with Red on top and Green at the bottom
- Image of size 200 x 200 with Red on top and Blue at the bottom

Yes, The answer is correct
Score : 1

Accepted Answer

Image of size 200 x 200 with Red on top and Blue at the bottom

---

10) Consider the program to estimate the area calcuation of your state. How can you increase the accuracy of the estimate? ***1 point***

Assignment 9

- By decreasing the number of iterations the experiment is performed.
- By reducing the number of iterations the experiment to one.
- By increasing the number of iterations to 100.

Yes, The answer is correct
Score : 1

Accepted Answer

By increasing the number of iterations the experiment is performed.

Assignment 10

Assessment due date is 25/11/2020 23:59:0

Last recorded submission : 2020-11-17, 12:53 IST

NOTE: Python 3.7 has been used for this Assignment

1) Which of the following attributes is used to obtain picture size using PIL in Python? ***1 point***

- len
- size
- dimension
- length

Yes, The answer is correct

Score : 1

Accepted Answer

size

---

2) Predict the output. ***1 point***

Assignment 10

```
l2=[1,4,6,12]
l=[]
for i in l1:
        for j in l2:
                if i==j:
                        l.append(i)

print(l)
```

- 1,4
- [1, 4, 6, 12]
- [1, 4]
- [2, 3, 5, 6, 12]

Yes, The answer is correct
Score : 1

Accepted Answer

[1, 4]

---

3) What is the output for the following snippet? ***1 point***

```
import numpy as np
m=np.array([[9,10,11],[19,20,21]])
print(m.T)
```

Assignment 10

```
[[ 9 19 10 20]]
```

```
[[ 9 19 11]
[10 20 21]]
```

```
[[ 9 19]
[10 20]
[11 21]]
```

Yes, The answer is correct
Score : 1

Accepted Answer

```
[[  9  19]
 [10  20]
 [11  21]]
```

4) What does the following command perform? ***1 point***

```
print('Chennai and Madras are same. Madras and Chennai are same.
    Chennai=Madras, Madras=Chennai'.replace('Madras','Chennai'
    ,2))
```

- Chennai and Chennai are same. Chennai and Chennai are same. Chennai=Chennai, Chennai=Chennai
- Chennai and Chennai are same. Chennai and Chennai are same. Chennai=Madras, Madras=Chennai

Assignment 10

- Chennai and Chennai are same. Chennai and Chennai are same. Chennai=Chennai, Madras=Chennai

Yes, The answer is correct
Score : 1

Accepted Answer

Chennai and Chennai are same. Chennai and Chennai are same. Chennai=Madras, Madras=Chennai

---

5) Given: ***1 point***

What is the output for the following command?

- array([5, 7, 9])
- array([6, 15])
- AxisError: axis 2 is out of bounds for array of dimension 2
- array([15])

Yes, The answer is correct
Score : 1

Accepted Answer

AxisError: axis 2 is out of bounds for array of dimension 2

---

6) Pick out the correct output. ***1 point***

```
array([[1., 1., 1.],
       [1., 1., 1.],
       [1., 1., 1.]])
```

Yes, The answer is correct
Score : 1

Accepted Answer

```
array([[1., 1., 1.],
       [1., 1., 1.],
       [1., 1., 1.]])
```

---

7) How will you create a blank image of the same size and type of a given image? ***1 point***

- img=Image.new(im.mode,im.shape)
- img=Image.new(im.mode,im.size)

img=Image.blank(im.mode,im.size)

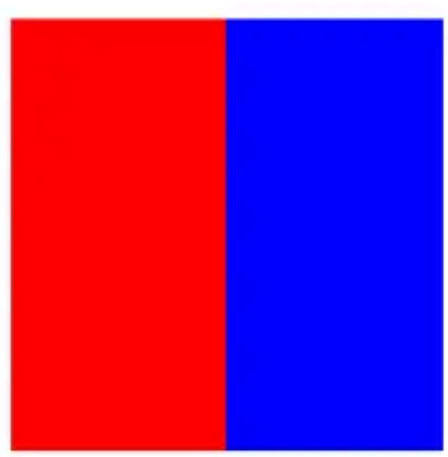
Yes, The answer is correct
Score : 1

Accepted Answer

img=Image.new(im.mode,im.size)

---

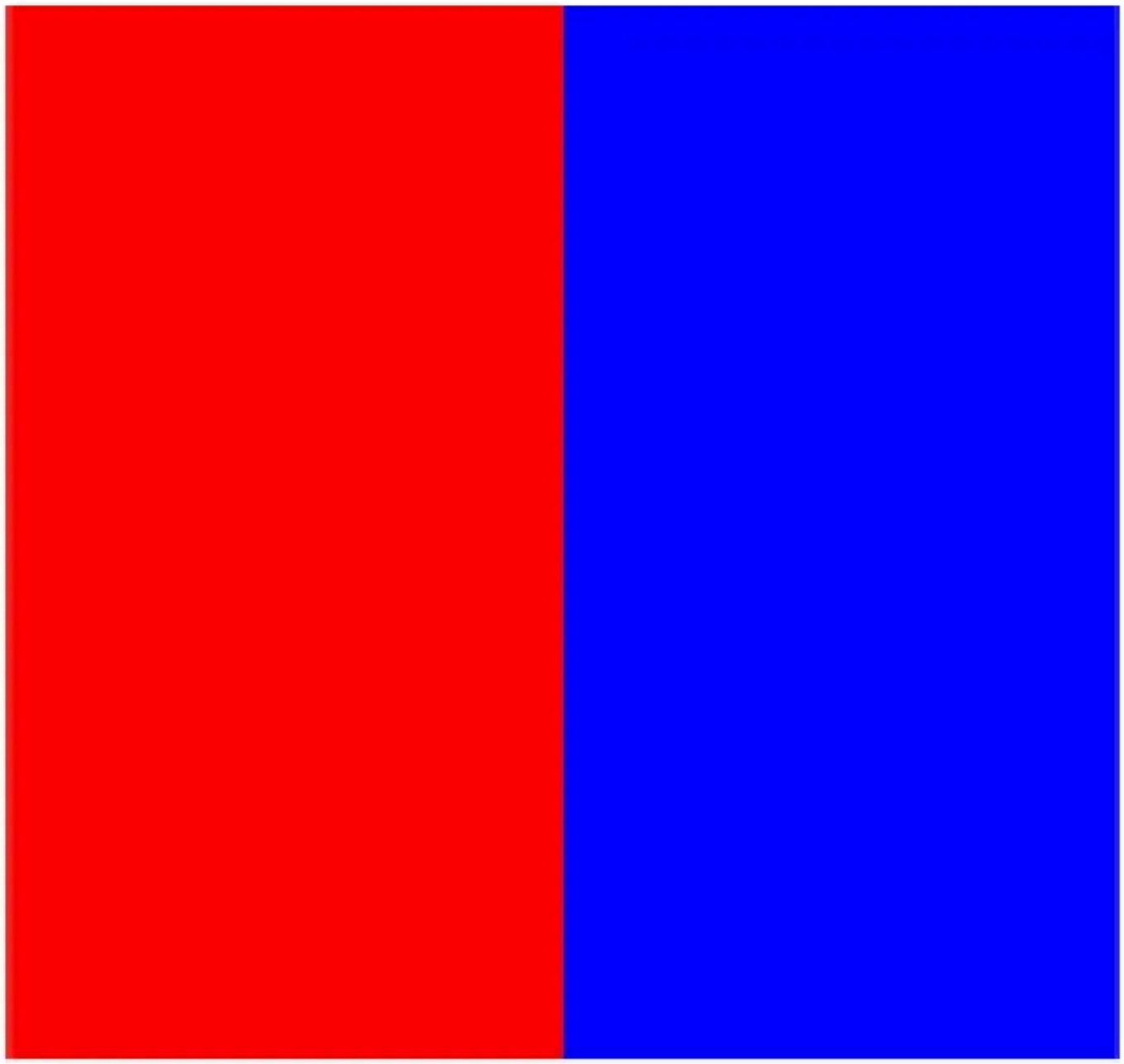
8) Choose the relevant image for test3.png created by the given code. ***1 point***



Yes, The answer is correct
Score : 1

Accepted Answer

Assignment 10



9) Predict the output. ***1 point***

- (•) 13
- ( ) 38

- 39

Yes, The answer is correct
Score : 1

Accepted Answer

13

---

10) Which of the following options is a best description of the output for the following command? ***1 point***

- 2 x 5 numpy array with random integers from 1 to 5
- 1 x 5 numpy array with random integers from 2 to 4
- 2 x 5 numpy array with random integers from 1 to 4
- 1 x 5 numpy array with random integers from 2 to 5

Yes, The answer is correct
Score : 1

Accepted Answer

2 x 5 numpy array with random integers from 1 to 4

Assignment 11

Assessment due date is 2/12/2020 23:59:0

Last recorded submission : 2020-11-28, 12:18 IST

NOTE: Python 3.7 has been used for this Assignment

1) Which of the following methods cannot be used to identify an element on the web page? ***1 point***

- find_element_by_id()
- find_element_by_link_text()
- find_element_by_class_name()
- send_keys()

Yes, The answer is correct
Score : 1

Accepted Answer

send_keys()

---

2) Identify the python library required for Browser Automation. ***1 point***

- Selenium
- Time
- Webdriver
- Chromedriver

Yes, The answer is correct
Score : 1

Assignment 11

3) Pick out the driver method to open a website in selenium. ***1 point***

- driver.getwebpage()
- driver.get()
- driver.request()
- driver.open()

Yes, The answer is correct
Score : 1

Accepted Answer

driver.get()

4) What is the purpose of the given command when we request to open a web page? ***1 point***

$$wait=WebDriverWait(driver,600)$$

- To record login information
- To check certificate information of the web page
- To give buffer time for the driver in case there is a slow network connection
- To collect cookies information

Assignment 11

Accepted Answer

To give buffer time for the driver in case there is a slow network connection

---

5) In Browser automation, you have identified the login box. Identify the instruction to type your name and press ENTER. ***1 point***

- [x] login_box.send_keys(name + Keys.ENTER)
- [ ] login_box.press_keys(name + Keys.ENTER)
- [ ] login_box.send_keys(name + ENTER)
- [ ] login_box.hot_key(name + Keys.ENTER)

Yes, The answer is correct
Score : 1

Accepted Answer

login_box.send_keys(name + Keys.ENTER)

---

6) Which of the following libraries can be used to print time according to different timezones? ***1 point***

- [ ] datetime
- [ ] date
- [ ] calendar
- [x] pytz

Yes, The answer is correct
Score : 1

Assignment 11

7) How will you find the day of the week, given a date? ***0 points***

```
import calendar
week_days=['Sunday','Monday','Tuesday','Wednesday','
    Thursday','Friday','Saturday']
i=calendar.weekday(2006,12,11)
print(week_days[i])
```

Yes, The answer is correct
Score : 0

Accepted Answer

```
import calendar
week_days=['Sunday','Monday','Tuesday','Wednesday','
    Thursday','Friday','Saturday']
i=calendar.weekday(2006,12,11)
print(week_days[i])
```

8) Find the output: ***1 point***

- month=11
- month=23
- month=2

Yes, The answer is correct
Score : 1

Accepted Answer

month=10

9) What is the last argument for the datetime function in the following command? ***0 points***

- nanoseconds
- milliseconds
- minutes
- microseconds

Yes, The answer is correct
Score : 0

Accepted Answer

microseconds

10) How will you retrieve current date using date library? ***1 point***

Assignment 11

- date.now()
- date.current date()
- date.date()

Yes, The answer is correct
Score : 1

Accepted Answer

date.today()

Assignment 12

Assessment due date is 9/12/2020 23:59:0

Last recorded submission : 2020-12-05, 12:27 IST

NOTE: Python 3.7 has been used for this Assignment

1) In a page rank algorithm, after taking an optimum number of random walks in a web graph, what can you say about the nodes with maximum points? ***1 point***

- (•) These nodes are the most visited
- ( ) These nodes are least visited
- ( ) These nodes have maximum number of in-links
- ( ) These nodes have maximum number of out-links

Yes, The answer is correct
Score : 1

Accepted Answer

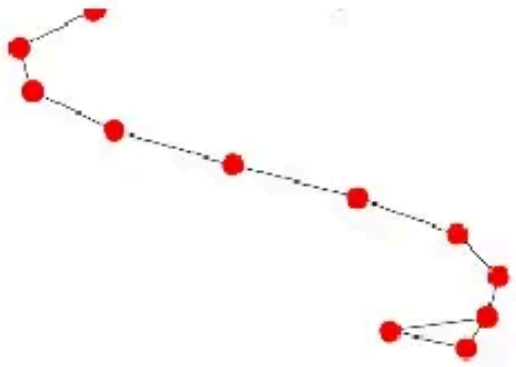
These nodes are the most visited

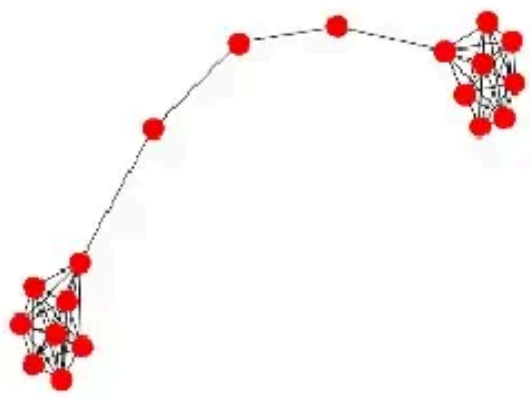
---

2) Identify the graph created using the following command. ***1 point***

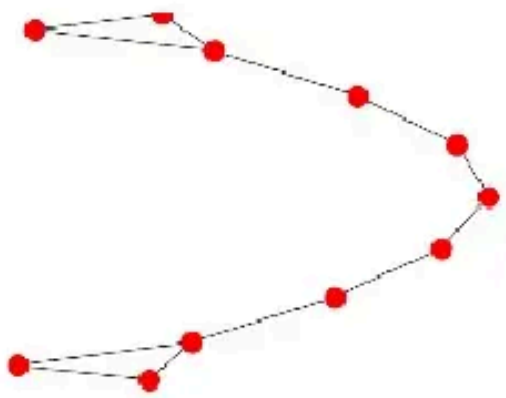
```
G = networkx.barbell_graph(8,3)
```

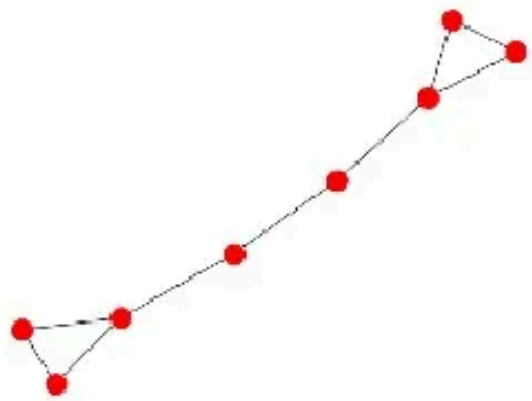
Assignment 12





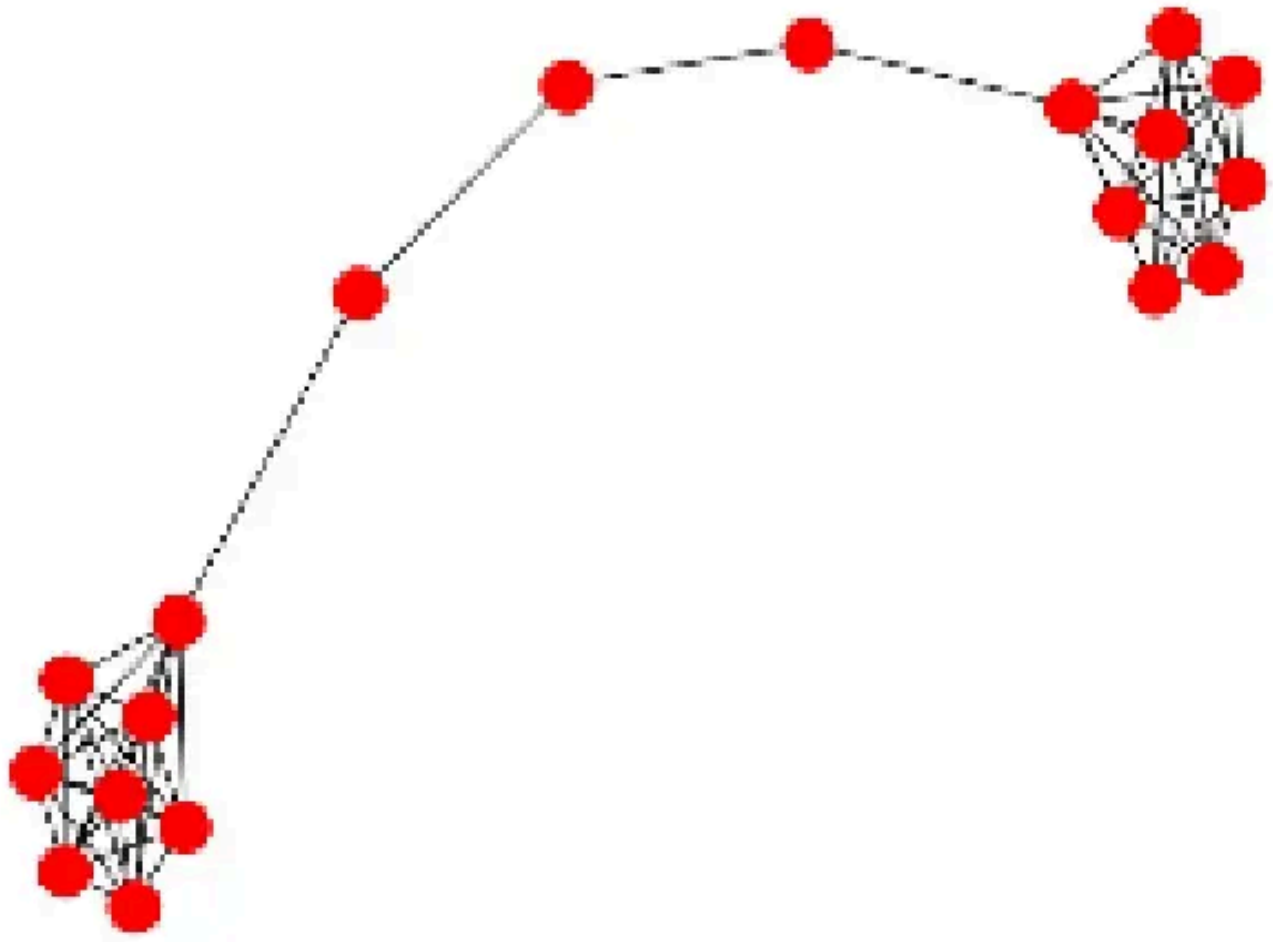
Assignment 12





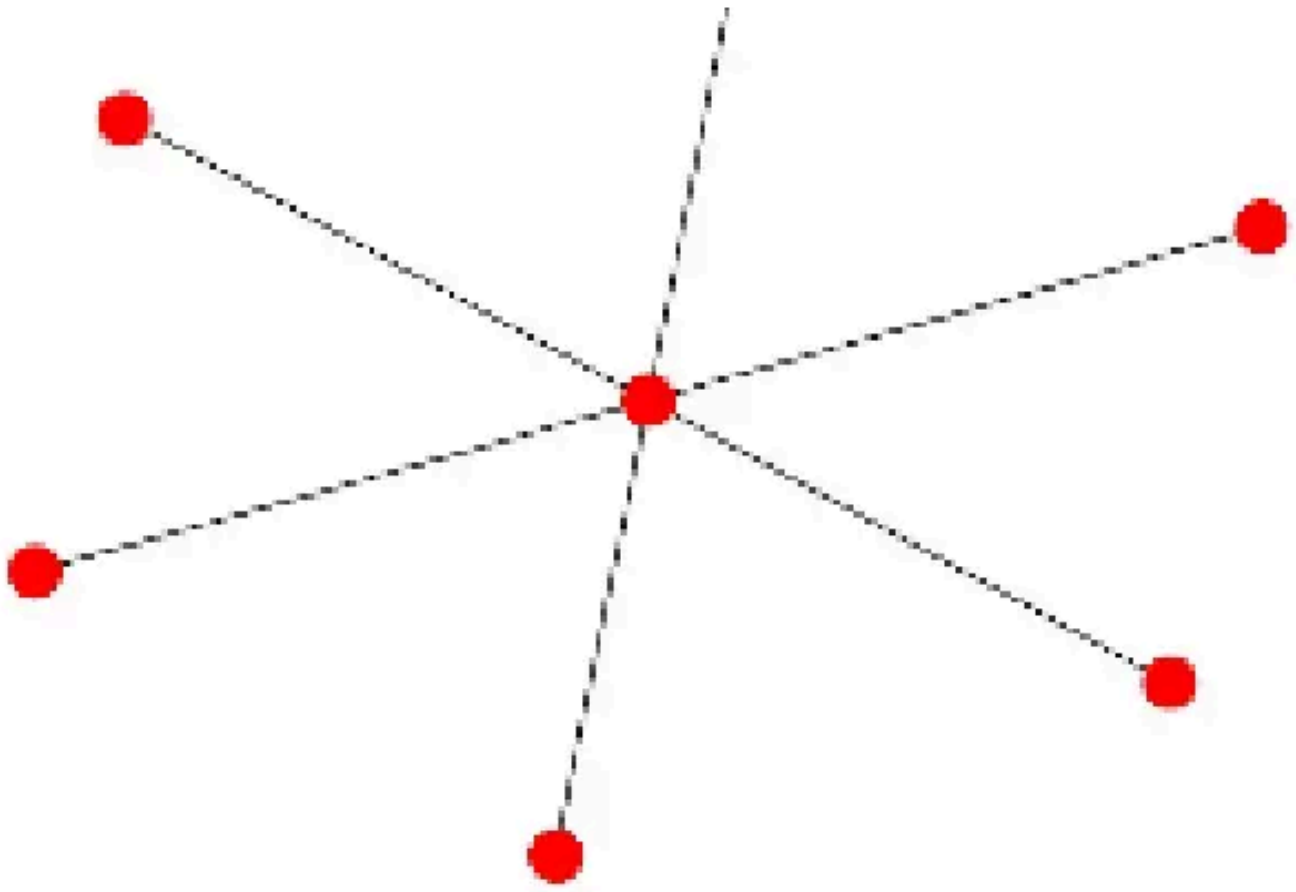
Yes, The answer is correct
Score : 1

Accepted Answer



3) Which of the following commands is used to create the graph given below: ***1 point***

Assignment 12



- G = networkx.complete_graph(7)
- G = networkx.cycle_graph(6)
- G = networkx.star_graph(6)
- G = networkx.star_graph(7)

Yes, The answer is correct
Score : 1

Accepted Answer

G = networkx.star_graph(6)

---

4) Comment on the purpose of the following command. ***1 point***

Assignment 12

- Sort the items of dictionary, p by key
- Sort the items of dictionary, p by values
- Sort the elements of list, p by values
- Sort the items of Tuple, p by values

Yes, The answer is correct
Score : 1

Accepted Answer

Sort the items of dictionary, p by values

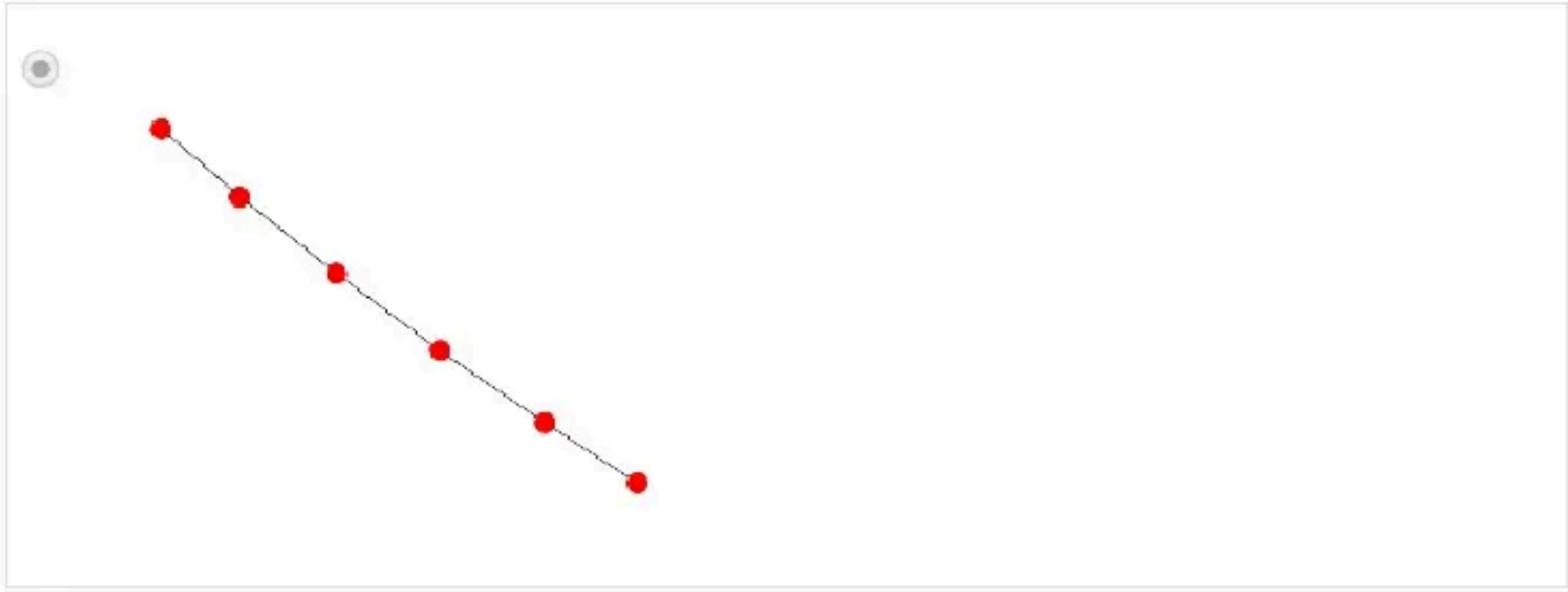
5) Identify the graph that can never be an output of the following code. ***1 point***

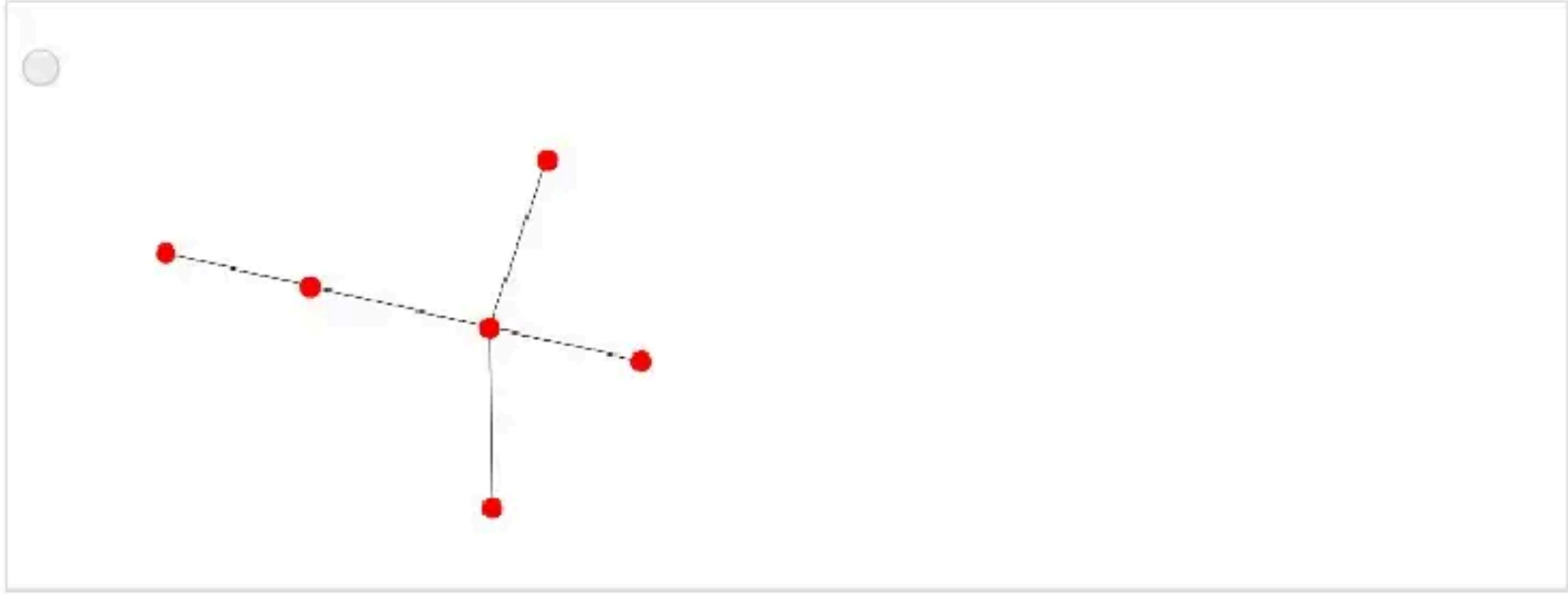
```
import random
import networkx as nx
import matplotlib.pyplot as plt

G = nx.Graph()
G.add_nodes_from([i for i in range(6)])
while (nx.is_connected(G)!=True):
        x=random.choice(list(G.nodes()))
        y=random.choice(list(G.nodes()))
        if (x!=y):
                G.add_edge(x,y)
```

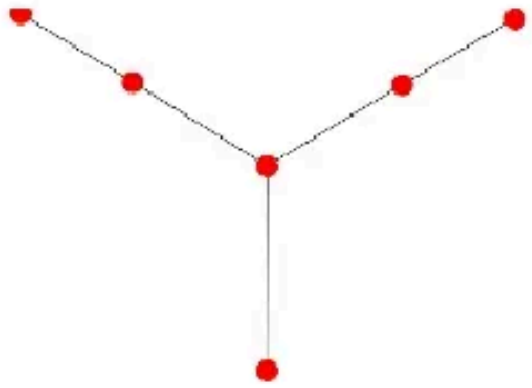
```
            else:
                continue

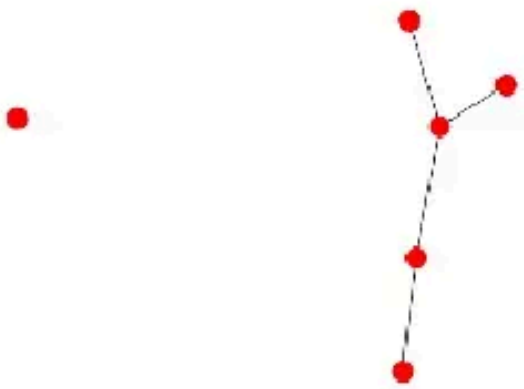
nx.draw(G)
plt.show()
```





Assignment 12

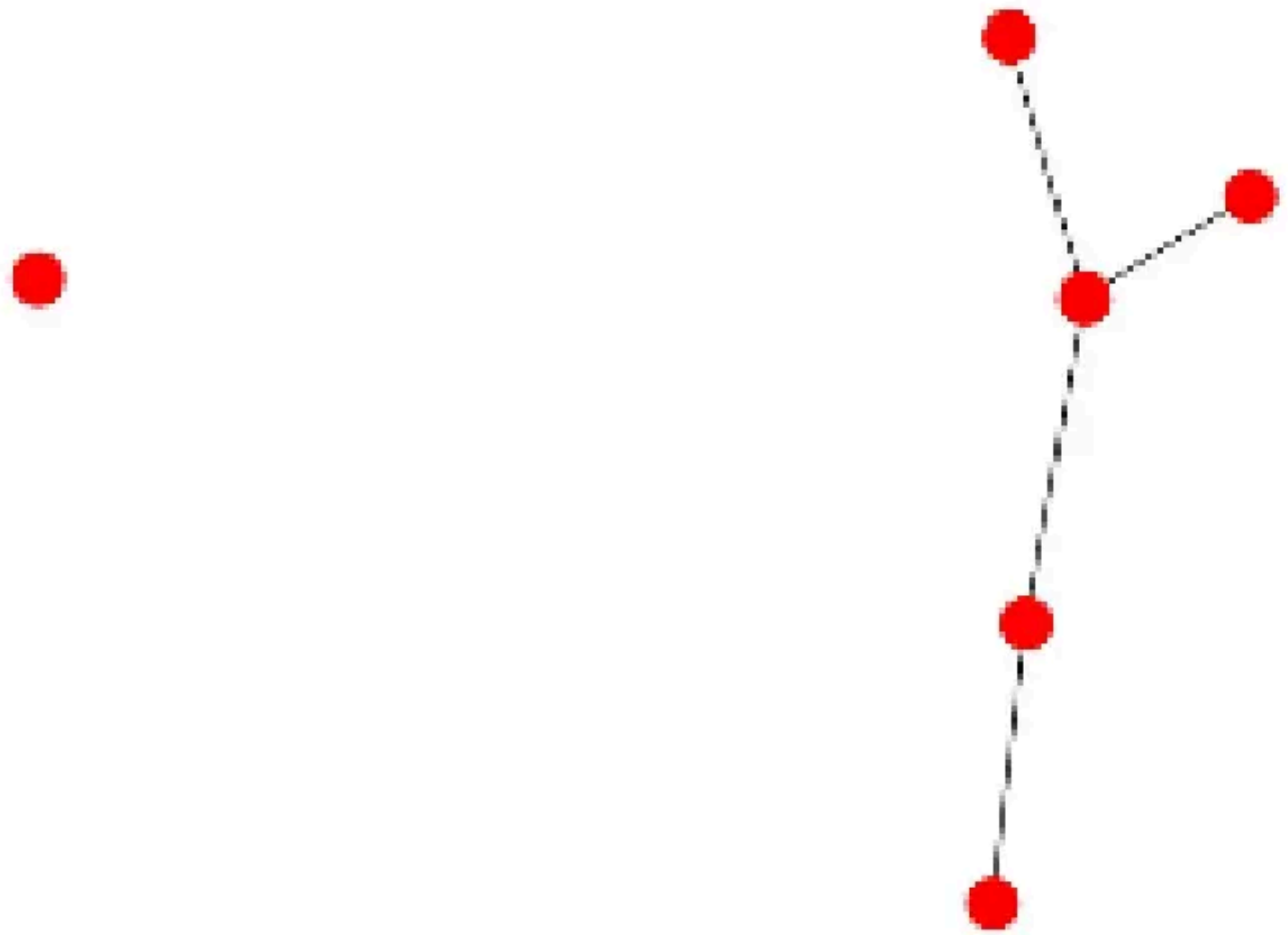




No, The answer is incorrect
Score : 0

Accepted Answer

Assignment 12



6) Which of the following real world networks represent an undirected graph? ***1 point***

- Wikigraph
- Citation network
- Facebook Friendship network
- World wide web

Yes, The answer is correct
Score : 1

Assignment 12

7) What happens to the total Points with every iteration in the Points distribution method? ***1 point***

- Increases every iteration
- Decreases for every iteration
- Increases or decreases depending on the structure of the graph
- Remains constant

Yes, The answer is correct
Score : 1

Accepted Answer

Remains constant

8) How many iterations does the number, 75 take to converge in Collatz Conjecture? ***0 points***

- 14
- 15
- 13
- 12

Yes, The answer is correct
Score : 0

Accepted Answer

14

Assignment 12

9) Which of the sequence do you obtain by executing 3n+1 algorithm for n = 10? ***1 point***

- 31, 15, 7, 3, 1
- 5, 16, 8, 4, 2, 1
- 5, 3, 2, 1
- doesnot converge

Yes, The answer is correct
Score : 1

Accepted Answer

5, 16, 8, 4, 2, 1

---

10) How will you choose the next node to traverse in a random walk method of Page Rank algorithm? ***1 point***

- Choose the node that has not been traversed yet
- Choose the most weighted out-link
- Randomly choose one of the out-links
- Choose the least weighted out-link

Yes, The answer is correct
Score : 1

Accepted Answer

Randomly choose one of the out-links

Assignment 12